

Predicting Diabetes Outcomes Using CDC BRFSS Data

Machine Learning for Population Health Classification

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Abstract

Diabetes affects over 37 million Americans and costs the healthcare system upward of \$327 billion annually. Early, accurate identification of at-risk individuals is one of the most direct levers available to reduce that burden, but most clinical screening tools still rely on individual physician judgment rather than systematic population-level signals. This project asks a simple question: how well can publicly available, self-reported health survey data predict a diabetes diagnosis, and where does that approach break down?

Using the CDC Behavioral Risk Factor Surveillance System (BRFSS) dataset (over 250,000 respondents), we trained and compared seven model families: Logistic Regression, XGBoost, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors, Naive Bayes, and K-Means clustering. Across the board, every supervised model converged on the same performance ceiling: an F1 score around 0.44 and a ROC-AUC around 0.82. Models were reasonably good at flagging people who *might* have diabetes (recall approx. 0.77 to 0.80) but generated a high rate of false positives (precision approx. 0.30 to 0.32). More critically, none of the models could meaningfully detect prediabetes, the one stage where intervention actually prevents progression to full diabetes.

The core finding is not a modeling failure. It is a data failure. The BRFSS dataset captures lifestyle and self-reported health status, but it does not include the clinical biomarkers (HbA1c, fasting glucose) that clinicians actually use to distinguish prediabetes from normal health. No amount of model complexity can substitute for missing features. The recommended next step is not a better algorithm; it is better data.

Overview

Problem statement. Undiagnosed diabetes and prediabetes represent a silent public health crisis. The CDC estimates that 96 million U.S. adults have prediabetes, and more than 80% of them are unaware of it. If identified early, lifestyle interventions reduce progression to type 2 diabetes by up to 58%. The practical question is whether a machine learning model trained on survey data can reliably surface high-risk individuals before they receive a clinical diagnosis.

Relevant literature. Prior work on health survey datasets has consistently found that self-reported features like BMI, blood pressure status, and general health self-assessment carry predictive signal for metabolic conditions. However, studies using clinical datasets with laboratory values (HbA1c, fasting glucose) achieve substantially higher precision, typically F1 scores above 0.65, because the clinical boundary for prediabetes is defined by those exact values. Survey-based models operate without that anchor.

Methodology. We approached this as both a binary classification problem (diabetes vs. no diabetes) and a multiclass problem (no diabetes / prediabetes / diabetes). Models were evaluated using stratified 80/20 train-test splits with cross-validation, class weighting, and SMOTE oversampling to address severe class imbalance. Feature engineering included interaction terms and composite risk scores, detailed in the Model Training section. Unsupervised K-Means clustering was used for exploratory analysis of natural risk groupings.

Data Processing

The BRFSS dataset contains 253,680 survey responses collected across the United States, with 21 health and lifestyle features per respondent. The binary classification target distinguishes diabetic from non-diabetic respondents. The multiclass target adds a prediabetes label that represents only 1.82% of the total dataset, a severe imbalance that shaped every preprocessing decision.

Key pipeline steps:

- **Stratified 80/20 split:** preserves class distribution across train and test sets, critical given minority class size.
- **Feature scaling:** z-score normalization applied to the three continuous features (BMI, MentHlth, PhysHlth), fit on training data only and applied to test sets using training parameters. Binary and ordinal features were left on their original scales. Scaling was required for Logistic Regression and SVM; XGBoost and Random Forest received the same scaled inputs for consistency, though tree-based models are scale-invariant.
- **Class imbalance handling:** three strategies were applied across models. Inverse-frequency class weights were used in Logistic Regression, Random Forest, and SVM, penalizing misclassification of the minority class at training time. XGBoost used `scale_pos_weight` (ratio of negative to positive class counts, approx. 6.2) for the binary task, and inverse-frequency sample weights passed into the DMatrix for the multiclass task. Both are native XGBoost approaches equivalent to class weighting. SMOTE (Synthetic Minority Oversampling Technique) was applied as a third approach, but strictly inside cross-validation folds so that synthetic samples never contaminate the validation fold.
- **Assumptions:** survey responses treated as accurate self-reports. No imputation was required; the BRFSS dataset has negligible missingness by design.

Key data limitation. The dataset contains no laboratory values. Features like CholCheck (whether the respondent had a cholesterol screening) serve as proxies for healthcare access and general health behavior, not as clinical indicators. This structural gap directly caps model precision throughout the project.

Data Analysis

Before fitting any predictive model, we ran exploratory analysis to understand the data's structure and whether the classes were separable at all. The short answer is that they are not, at least not with these features.

PCA: class overlap.

Principal Component Analysis reduced the dataset to two dimensions to visualize class boundaries. The first principal component (PC1) captures 46.4% of total variance and is driven primarily by Age (loading 0.97), with secondary contributions from Income, GenHlth, and HighBP. Despite this strong first component, 2D PCA plots for both binary and multiclass datasets show extensive overlap between class clouds. There are no clean decision boundaries. This is the clearest early signal that linear separability is not achievable with these features.

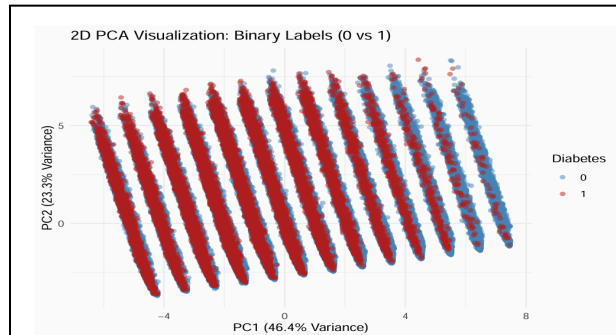


Figure 1. PCA projection (binary). True class labels (0 = no diabetes, 1 = diabetes) show complete overlap across the feature space. No separable boundary exists..

K-Means: risk stratification, not classification.

K-Means clustering ($k = 3$, selected via elbow method on within-cluster sum of squares, which showed diminishing returns beyond $k = 4$) produced three interpretable risk groups visible in Figure 2: a high-risk cluster with elevated BMI, poor general health, and ~21% diabetes rate; a low-risk cluster with opposite characteristics and ~10% diabetes rate; and a moderate-risk intermediate group. The overlap between clusters along PC1 is visible in the plot and confirmed quantitatively by Adjusted Rand Index values of 0.02 to 0.07, indicating these groupings do not map onto actual class labels. K-Means is useful for population segmentation but cannot substitute for supervised classification.

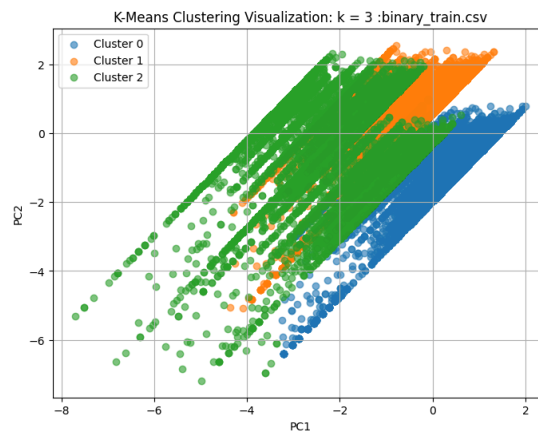


Figure 2. K-Means clustering ($k = 3$, binary dataset). Three risk groups emerge along PC1, but cluster boundaries overlap substantially, confirming that natural groupings do not align with true class labels.

Feature distributions.

CholCheck (whether the respondent received a cholesterol screening) shows the strongest bivariate association with the diabetes label. High blood pressure, high cholesterol, BMI, and self-rated general health (GenHlth) follow. These features reflect broad health burdens rather than clinical diabetes etiology; they tell us someone is generally unhealthy, not specifically diabetic.

Model Training

We trained seven model families and evaluated each on both binary and multiclass tasks. Logistic Regression was established as the interpretable linear baseline. Naive Bayes was included as a probabilistic baseline, offering a fast, low-assumption benchmark grounded in feature independence. XGBoost and Random Forest were used to test whether nonlinear ensemble methods could break through the baseline ceiling. SVM with RBF kernel explored nonlinear decision boundaries with a different inductive bias. K-Nearest Neighbors (KNN) was evaluated as an instance-based, non-parametric alternative, with K tuned across values of 3 to 15 via sensitivity analysis. K-Means served as an unsupervised complement.

Feature engineering (XGBoost).

Beyond raw features, we constructed six engineered features for the binary task, including interaction terms (BMI_Age, HighBP_HighChol), a composite cardiovascular risk score (MetabolicRisk), and binary indicators for poor physical health days, poor mental health days, and sedentary obesity. For interpretability analysis, we focused on the two features with the strongest SHAP signal. GenHlth remained the top predictor (SHAP = 0.64), but BMI_Age jumped to second (SHAP = 0.45) and MetabolicRisk to third (SHAP = 0.42), together absorbing the standalone Age signal (which dropped to 0.055). None of the engineered features improved test performance. XGBoost was already capturing these interactions internally. Feature engineering improved interpretability without moving the needle on accuracy.

Evaluation metrics.

Primary metrics: F1 score (harmonic mean of precision and recall, appropriate for imbalanced classes) and ROC-AUC (overall discriminative ability across thresholds). We tracked precision and recall separately because the clinical stakes are asymmetric: missing a diabetic patient (false negative) has different consequences than flagging a healthy one (false positive). Cross-validation F1 was the selection criterion within each model family.

Model selection rationale.

Within Logistic Regression, all three binary variants (class weights, SMOTE, Ridge with class weights) converged to nearly identical CV F1 around 0.442 and CV AUC around 0.822, so class weights were selected for the binary task for simplicity. For the multiclass task, Multinomial Ridge with SMOTE was selected over One-vs-Rest with class weights based on higher CV macro F1 (0.4595 vs 0.4379); Multinomial Ridge models all three classes simultaneously rather than treating each as an independent binary problem. For XGBoost, the raw feature set performed marginally better on the binary task, but the engineered variant was selected for both tasks to enable richer SHAP analysis without meaningful performance cost. XGBoost hyperparameters were tuned via grid search over max_depth, eta, min_child_weight, subsample, colsample_bytree, and gamma on a stratified 90/10 validation slice, with early stopping on val AUC. Random Forest used 500 trees, max depth 15, and inverse-frequency class weights for both tasks. SVM used RBF kernel with class weighting as the primary approach, with SMOTE inside CV folds as an alternative.

Model Validation

All models were evaluated on a held-out test set, with cross-validation results used to confirm that test performance was not an artifact of a lucky split. Across every model, CV and test metrics aligned closely and there is no meaningful overfitting.

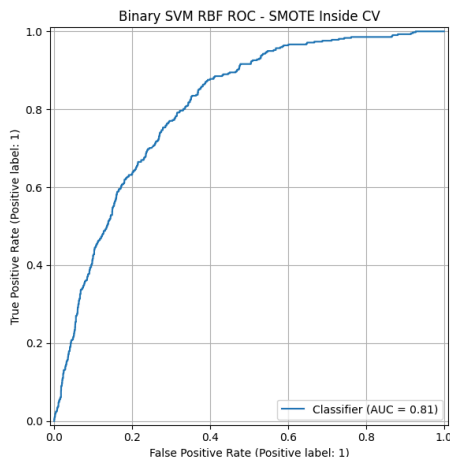


Figure 3. ROC curve, binary SVM with SMOTE. AUC = 0.82 is representative of all models.

Binary task.

Every supervised model landed in a tight band: F1 $\approx$ 0.44, ROC-AUC $\approx$ 0.82, recall approx. 0.75 to 0.80, precision approx. 0.30 to 0.32. High recall means models catch most actual diabetes cases. Low precision means roughly two out of every three positive predictions are false alarms. For a population-level screening tool, high recall is the right priority, but the false positive rate is high enough that downstream clinical resources would be strained. Naive Bayes and KNN resulted in the lowest recalls, though KNN at k = 15 had the highest precision of the models performed.

Multiclass task.

Performance degraded substantially when prediabetes was added as a third class. Overall accuracy ran 0.65 to 0.79, but macro F1 dropped to 0.41 to 0.44 because prediabetes (Class 1) was effectively undetectable across all models, with F1 scores of 0.02 to 0.06, precision as low as 0.03. KNN achieved high accuracy, however, nearly zero F1 score for prediabetes. The model either ignored the class entirely or assigned it at random. Even aggressive SMOTE and class weighting did not fix this. The prediabetes label does not have a distinct, learnable footprint in self-reported survey data.

Bias and risk.

The most practically significant bias finding came from age stratification in the Random Forest: recall for respondents in younger age brackets (18 to 34) was between 0% and 16%, while recall for respondents 65 and older exceeded 81%. The model has learned that diabetes is an older person's disease, which is statistically true in this dataset, but it fails almost entirely for younger diabetics. Any deployment of this model as a screening tool would produce systematically worse outcomes for younger populations.

Model Performance

Binary classification results across all models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.731	0.312	0.769	0.444	0.824
XGBoost	0.719	0.305	0.799	0.442	0.832
Random Forest	0.751	0.327	0.743	0.454	0.828
SVM (class weights)	0.710	0.300	0.800	0.440	0.820
SVM (SMOTE in CV)	0.720	0.300	0.750	0.440	0.810
K-Nearest Neighbors (k = 15)	0.787	0.343	0.577	0.431	0.821
Naive Bayes	0.741	0.305	0.667	0.418	0.753

Table 1. Binary classification: all models converge to the same performance ceiling.

Multiclass results (macro-averaged), selected models:

Model	Accuracy	Macro F1	Prediabetes F1	Avg AUC
Logistic Regression (Ridge + SMOTE)	0.650	0.426	0.056	0.770
XGBoost (engineered features)	0.788	0.440	0.022	0.741
Random Forest	0.714	0.437	0.043	~0.760
SVM RBF (SMOTE in CV)	0.720	0.410	0.050	~0.740
K-Nearest Neighbors (k = 7)	0.835	0.596	0	X
Naive Bayes	0.755	0.430	0.0320	X

Table 2. Multiclass classification: prediabetes F1 collapses across all models.

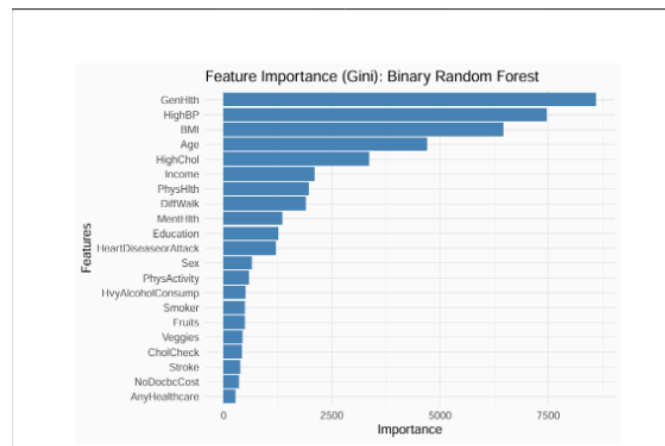


Figure 4. Random Forest feature importance (binary). GenHlth and HighBP dominate; CholCheck is strongest in Logistic Regression.

The key observation across both tables is that increasing model complexity, from logistic regression to gradient boosting to deep ensembles, produces no meaningful gain. Random Forest's F1 of 0.454 is marginally better than Logistic Regression's 0.444, but not by an amount that would matter operationally. The performance ceiling is flat because the features are flat. CholCheck, HighBP, HighChol, BMI, and GenHlth are the top predictors across every model family, not because they are strong clinical signals, but because they are the best proxies available in a dataset that has no clinical signals.

Conclusion

This project set out to predict diabetes outcomes from population health survey data. The models work, in the sense that a ROC-AUC of 0.82 is meaningfully better than random, and a recall of 0.77 to 0.80 means that a real screening program based on these models would catch most diabetic individuals. But they do not work well enough to be clinically useful as-is, and they fail entirely on the one task that matters most: identifying prediabetes before it becomes diabetes.

What went right. Binary classification is consistent and reproducible across all seven model families. The models agree on which features matter. The K-Means risk stratification cleanly separates high-risk from low-risk population segments, which could be useful for public health targeting even if it cannot make individual-level diagnoses.

What went wrong. Precision is too low for clinical deployment. A 70% false positive rate would overwhelm follow-up resources. Prediabetes detection is essentially zero. The age bias is severe enough that deploying this model would actively worsen health equity outcomes for younger populations.

Root cause. These are not model failures. No model architecture will fix a dataset that lacks HbA1c and fasting glucose values. Prediabetes is defined clinically by HbA1c between 5.7% and 6.4%, or fasting glucose between 100 and 125 mg/dL. Without those numbers, the model has no basis for placing the prediabetes boundary, and all the SMOTE and class weighting in the world cannot manufacture a signal that is not in the data.

Recommendations. First, integrate even a small subsample of clinical biomarkers (HbA1c, fasting glucose) and re-run the binary models. This single change is likely to increase precision substantially and make prediabetes detectable. Second, address the age bias explicitly before any deployment by training age-stratified models or applying post-hoc calibration. Third, when clinical biomarkers are available, the binary model is the more viable deployment candidate than the multiclass one. A high-recall screening tool that flags at-risk individuals for follow-up testing is a realistic application of this work. A tool that attempts to diagnose prediabetes without clinical biomarkers is not ready for deployment.

Caution. Survey self-report bias is real. The apparent 'protective' effect of heavy alcohol consumption in the Logistic Regression odds ratios ($OR < 1$) is almost certainly a survivorship artifact. Heavy drinkers with diabetes complications may underreport or drop out of the survey. Any production deployment must account for the fact that the training population is a self-selected survey respondent, not a patient population.

Data Sources

- **CDC BRFSS Dataset (UCI Repository):** <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>
- **CDC BRFSS Dataset (Kaggle mirror, Teboul 2022):**
<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>
- **CDC MMWR Diabetes Trends Report:** <https://www.cdc.gov/mmwr>, Burrows et al. 2017, Vol. 66(43)

Source Code

Python libraries: scikit-learn, pandas, numpy, imbalanced-learn (SMOTE), matplotlib, shap

R libraries: ranger (Random Forest), glm / glmnet (Logistic Regression), xgboost, cluster, factoextra (K-Means, PCA)

Code covers: data preprocessing and scaling, stratified train/test splitting, model training with cross-validation, SMOTE within CV folds, SHAP feature importance, ROC/AUC computation, confusion matrices, PCA visualization, K-Means elbow plots and ARI scoring, age-stratified recall analysis.

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